

Brad Phelan — CV

Brad Phelan

Principal Software Engineer

Aachen, Germany · github.com/bradphelan

SUMMARY

Principal engineer with 25+ years of experience across embedded systems, desktop software, and geometric algorithms. Deep expertise in CAD/CAM toolpath generation and computational geometry, combined with a track record of leading small, high-output engineering teams. Comfortable at every layer of the stack — from safety-critical embedded firmware to functional language design, from C++ algorithm architecture to full-stack web. Currently driving AI tooling adoption and developer infrastructure strategy at ModuleWorks.

EXPERIENCE

Principal Software Engineer

ModuleWorks · Full-time

April 2019 – Present · Aachen, Germany

ModuleWorks is the world's leading developer of CAD/CAM software components for CNC machine control and simulation, holding over 70% global market share.

- Design and adapt machining algorithms in collaboration with product managers and domain specialists
 - Developed the Rotary Machining module, including axis-offset strategies and conical part machining, shipped in Autodesk Fusion 360
 - Extended the Turning component with improved roughing features, B-axis finishing cycles, and threading
 - Leading AI tooling strategy and developer infrastructure, including evaluation and rollout of LLM-assisted coding tools (Claude Code, Cursor) and participation in the AI Architecture Review Board
 - Working languages: C++, Python, CMake
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Lead Software Engineer — C# / .NET

Weingärtner Maschinenbau GmbH

August 2012 – April 2019 · 6 yrs 9 mos · Kirchham, Austria

Weingärtner designs and manufactures high-precision machine tools for the oil & gas and pump industries.

- Built weinCAD moineau: end-to-end CAD/CAM software for the design and manufacture of progressive cavity pumps and mud motors
 - Developed geometric algorithms for specialised 5-axis toolpath generation on non-standard workpiece geometry
 - Owned the full stack: geometric kernel, CAD UI (WPF), CAM engine, and machine post-processor
 - Working languages: C#, .NET, WPF, OpenGL, SolidWorks API
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Senior Software Engineer / Team Lead

mySugr Apps OG

March 2011 – December 2011 · 10 mos · Vienna, Austria

mySugr is a digital diabetes management platform (acquired by Roche).

- Led mobile engineering team building the mySugr diabetes logbook app
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Software Engineer

trikeapps

January 2011 – February 2011 · 2 mos · Melbourne, Australia

- Ruby on Rails contracting
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Software Engineer / Project Manager

Stream Unlimited

November 2007 – November 2010 · 3 yrs 1 mo · Vienna, Austria

Stream Unlimited is a specialist in embedded streaming and voice assistant technology.

- Built a personalised content recommendation engine for a personal video recorder: a Bayesian classifier that learns viewer preferences and surfaces relevant material; stack: C++, Python, Ruby, SQL
- Architected a high-availability version of the same system using Linux, MySQL, DRBD, Pacemaker, and AMQP
- Delivered the project as both engineer and project manager

Software Engineer

TTTech Computertechnik AG

November 2005 – November 2007 · 2 yrs 1 mo · Vienna, Austria

TTTech is a leader in safety-critical networking for aerospace and automotive.

- Embedded systems development for safety-critical aerospace software under DO-178B processes
- Contracting in MATLAB and Simulink model-based design

Contractor — MATLAB / Simulink

XTargets

April 2005 – October 2005 · 7 mos · Austria

- MATLAB and Simulink development contracts

Senior Engineer

The MathWorks

January 1999 – November 2003 · 4 yrs 11 mos · Cambridge, United Kingdom

- Principal technical engineer on the Embedded Target for MPC555, part of Real-Time Workshop and Embedded Coder
- Developed software for the embedded systems product line used in production code generation for automotive and aerospace

Software Engineer

Xerox

April 1998 – December 1998 · 9 mos · Welwyn Garden City, United Kingdom

- Embedded systems development for Xerox photocopier firmware

Graduate / Project Engineer

Bechtel

January 1998 – July 1998 · 7 mos · Boyne Island, Queensland, Australia

- Graduate engineer supporting commissioning of a new aluminium smelter at Boyne Island, Queensland

EDUCATION

University of Melbourne

Bachelor of Engineering (Electronics) · 1992 – 1997

University of Melbourne

Bachelor of Computer Science · 1992 – 1997

Activities: Theatre — actor and stage hand in university productions

SKILLS

Languages & Runtimes

C++, C#, Python, Ruby, Rust, SQL, MATLAB / Simulink

Domains

CAD/CAM · Computational Geometry · Toolpath Generation · Embedded Systems · Safety-Critical Software (DO-178B) · Machine Learning / Bayesian Inference · Algorithm Design · Linear Algebra

Frameworks & Tools

CMake · .NET / WPF / WinForms · OpenGL · SolidWorks API · Git · GitHub · Visual Studio · Linux · TCP/IP · Relational Databases · Agile

Practices

Software Architecture · Technical Leadership · AI Tooling Strategy · Project Management · Functional Programming · Asynchronous Systems

LANGUAGES

English — Native

German — Limited working proficiency

INTERESTS

Snowboarding · Wing foiling · Salsa & Bachata · Cooking · Visual art